Module 1: Stellarator Equilibrium and Magnetic Geometry

Textbook Physics Note, Version 1.0

From elementary magnetic confinement to three-dimensional flux coordinates and equilibrium data products

Purpose and status. This chapter develops the physics and mathematics needed to understand a stellarator equilibrium from the beginning. It assumes no prior plasma-physics background beyond elementary calculus and vectors. Every abbreviation is expanded at first use, every coordinate and field variable is defined, and the physical meaning of the principal equations is stated alongside the mathematics. The chapter explains what an equilibrium is, how magnetic surfaces and field lines are represented, what rotational transform and magnetic shear mean, why a stellarator requires a genuinely three-dimensional description, and which geometric quantities must be delivered to the remaining kinetic-magnetohydrodynamic modules. This is a completed educational physics note. It does *not* claim that a validated three-dimensional equilibrium solver has been implemented in this repository; the corresponding implementation note remains a solver plan.

Contents

How to use this chapter

A reader encountering plasma confinement for the first time should read Sections 1–3 in order. Sections 5–7 introduce the geometry that is required by Module 4, which computes Alfvén continua and eigenmodes. Sections 9–18 explain how three-dimensional stellarator equilibria are represented and computed. Sections 20–22 define the data products and verification obligations for the project.

The central dependency chain is

$$\begin{aligned}\text{coils and plasma profiles} &\longrightarrow \text{magnetohydrodynamic force balance}\\ &\longrightarrow \text{magnetic surfaces and field-line geometry}\\ &\longrightarrow \text{orbit, wave, and transport calculations.}\end{aligned} \tag{0.1}$$

The purpose of Module 1 is to establish the first three links reliably enough that every downstream calculation uses a common geometry and a common set of conventions.

Learning objectives. After completing this chapter, the reader should be able to:

1. explain why toroidal magnetic confinement requires helical field-line winding;
2. derive the static ideal-magnetohydrodynamic force-balance equation and interpret magnetic pressure and tension;
3. define magnetic surfaces, toroidal and poloidal flux, rotational transform, safety factor, and magnetic shear;
4. distinguish geometric angles from straight-field-line magnetic coordinates;
5. interpret three-dimensional stellarator Fourier coefficients and field periods;
6. describe fixed-boundary and free-boundary equilibrium calculations;
7. identify the geometry, derivatives, metrics, and metadata required by the downstream orbit, wave, transport, and surrogate modules; and
8. design verification tests that separate numerical convergence from physical validation.

Recommended prerequisite basics notes

This chapter is designed to be readable without first completing the entire basics curriculum. The following notes provide useful parallel explanations.

Basics note	Subject	Connection to Module 1
Basics 01	Units, dimensions, and normalization	Reviews the International System of Units, dimensional checks, characteristic scales, and normalized variables.
Basics 02	Vectors, fields, and coordinates	Introduces vector components, basis vectors, cylindrical coordinates, and curvilinear coordinates.
Basics 03	Vector calculus	Reviews gradient, divergence, curl, flux, and integral theorems.
Basics 11	Electric and magnetic fields	Introduces electromagnetic fields and their physical interpretation.
Basics 12	Maxwell equations	Gives the field equations from which the magnetostatic relations used here are obtained.
Basics 17	What is a plasma?	Explains quasineutrality, collective behavior, species, and plasma parameters.
Basics 21	Fluid descriptions	Explains how particle species are represented by density, velocity, and pressure fields.
Basics 24	Ideal magnetohydrodynamics	Derives the bulk equations whose time-independent limit defines the present equilibrium problem.
Basics 26	Ideal Ohm's law and flux freezing	Explains why magnetic field lines move with a highly conducting ideal plasma.
Basics 59	Magnetic-confinement concepts	Provides a shorter conceptual introduction to toroidal confinement.
Basics 60	Tokamaks and stellarators	Compares the two principal toroidal magnetic-confinement concepts.
Basics 61	Toroidal coordinates and magnetic surfaces	Develops the geometric language used in Sections 5–7.
Basics 62	Rotational transform and shear	Expands the field-line winding concepts used throughout the project.
Basics 63	Stellarator symmetry and Fourier geometry	Gives a focused account of the spectral surface representation used in Section 9.
Basics 65	Coils and the Biot–Savart law	Connects engineering coil currents to the vacuum magnetic field.

Basics note	Subject	Connection to Module 1
Basics 66	Equilibrium codes and VMEC concepts	Gives a companion introduction to numerical three-dimensional equilibrium solvers.

Table 1: Recommended supporting notes. Basics 59–66 are curriculum scaffolds in the present repository baseline; Module 1 supplies the complete first-pass explanation needed here.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
AE	Alfvén eigenmode	A coherent oscillation whose restoring force is largely magnetic tension; computed in Module 4.
Boozer coordinates	Straight-field-line magnetic coordinates	Flux coordinates in which magnetic-field representations are especially useful for particle orbits and neoclassical transport.
Cylindrical coordinates	(R, ϕ, Z)	R is distance from the symmetry axis, ϕ is the geometric toroidal angle, and Z is vertical position.
EP	Energetic particle	A particle population with energy well above the bulk thermal energy, such as fusion-born alpha particles or neutral-beam ions.
FEM	Finite-element method	A numerical method based on weak forms and piecewise basis functions.
FFT	Fast Fourier transform	An efficient algorithm for converting sampled periodic data to Fourier coefficients and back.
LCFS	Last closed flux surface	The outermost closed magnetic surface used to define the confined plasma region in a nested-surface model.
MHD	Magnetohydrodynamics	A fluid description in which a conducting plasma interacts with electric and magnetic fields.
NBI	Neutral beam injection	A heating and current-drive method that injects fast neutral atoms, which ionize inside the plasma.

Symbol / acronym	Name	Meaning in this document
PINN	Physics-informed neural network	A neural approximation constrained partly by differential-equation residuals and boundary conditions.
QA	Quasi-axisymmetry	A form of quasisymmetry in which magnetic-field strength is approximately axisymmetric in suitable magnetic coordinates.
QH	Quasi-helical symmetry	A form of quasisymmetry in which magnetic-field strength approximately follows a helical symmetry.
SI	International System of Units	The unit system used unless a normalized variable is explicitly introduced.
VMEC	Variational Moments Equilibrium Code	A widely used code that computes ideal-MHD equilibria with nested magnetic surfaces using a variational formulation.
$\mathbf{x}$	Position vector	A point in physical three-dimensional space, usually measured in metres.
(x, y, z)	Cartesian coordinates	Orthogonal coordinates used for general vector operations.
(R, ϕ, Z)	Cylindrical coordinates	Coordinates adapted to toroidal devices; ϕ increases around the major axis.
(s, θ, ζ)	Magnetic flux coordinates	s labels a magnetic surface, θ is a poloidal angle, and ζ is a toroidal angle.
R_0	Reference major radius	A characteristic distance from the device symmetry axis to the magnetic axis.
a	Reference minor radius	A characteristic cross-sectional radius of the plasma.
A	Aspect ratio	The dimensionless ratio $A = R_0/a$.
N_{fp}	Number of field periods	Number of repeated toroidal sectors in a stellarator magnetic configuration.
$\mathbf{B}$	Magnetic flux density	The magnetic field measured in tesla; $B = \lvert\mathbf{B}\rvert$ is its magnitude.
$\mathbf{b}$	Magnetic-field unit vector	$\mathbf{b} = \mathbf{B}/B$.
$\mathbf{E}$	Electric field	Force per unit charge, measured in volts per metre.
$\mathbf{J}$	Electric current density	Electric current per unit area, measured in amperes per square metre.

Symbol / acronym	Name	Meaning in this document
p	Scalar plasma pressure	Isotropic thermodynamic pressure, measured in pascals, in the ideal-MHD equilibrium model.
ρ_m	Mass density	Plasma mass per unit volume, measured in kilograms per cubic metre.
n_s	Number density of species s	Number of particles of species s per unit volume.
T_s	Temperature of species s	Thermal energy scale; when stated in kelvin it is multiplied by Boltzmann's constant in energy expressions.
q_s	Charge of species s	Electric charge of a particle species.
m_s	Mass of species s	Particle mass of species s.
μ_0	Vacuum permeability	Magnetic constant, approximately $4\pi \times 10^{-7}\ \mathrm{H\,m^{-1}}$.
ϵ_0	Vacuum permittivity	Electric constant in Maxwell's equations.
ψ	Toroidal-flux coordinate	A magnetic-surface label proportional to toroidal magnetic flux; the exact normalization must always be stated.
Ψ_t	Toroidal magnetic flux	Magnetic flux through a poloidal cross-section bounded by a magnetic surface.
Ψ_p	Poloidal magnetic flux	Magnetic flux through a toroidal ribbon bounded by a magnetic surface.
s	Normalized flux coordinate	Commonly $s = \Psi_t/\Psi_{t,a}$, where the subscript a denotes the plasma boundary.
ρ	Normalized radius	Often $\rho = \sqrt{s}$; this symbol is not the mass density ρ_m.
θ	Poloidal magnetic angle	Periodic angle that advances around a short way through the plasma cross-section.
ζ	Toroidal magnetic angle	Periodic angle that advances around the long way through the torus.
ι	Rotational transform	Average poloidal turns per toroidal turn made by a magnetic field line.
q	Safety factor	In the convention used here, $q = 1/\iota$; common in tokamak literature.
$\hat{s}$	Magnetic shear	A dimensionless measure of radial variation of field-line pitch; a specific definition is given in Section 6.

Symbol / acronym	Name	Meaning in this document
m	Poloidal mode number	Integer counting variation in the poloidal angle.
n	Toroidal mode number	Integer counting variation in the toroidal angle under the stated Fourier convention.
$\sqrt{g}$	Coordinate Jacobian	Volume scale factor satisfying $dV = \sqrt{g}\,ds\,d\theta\,d\zeta$.
g_{ij}	Covariant metric tensor	Dot products of covariant basis vectors: $g_{ij} = \partial_i\mathbf{x} \cdot \partial_j\mathbf{x}$.
g^{ij}	Contravariant metric tensor	Dot products of gradients: $g^{ij} = \nabla u^i \cdot \nabla u^j$, where u^i denotes a coordinate.
$V(s)$	Enclosed volume	Physical volume inside the surface labeled by s.
$V'(s)$	Differential volume	Derivative dV/ds, used in flux-surface transport equations.
$\langle A\rangle_s$	Flux-surface average	Surface-weighted average of a scalar A over the magnetic surface s.
$\boldsymbol{\kappa}$	Magnetic curvature	$\boldsymbol{\kappa} = \mathbf{b} \cdot \nabla\mathbf{b}$, directed toward the local center of curvature of a field line.
β	Plasma beta	Ratio of plasma pressure to magnetic pressure; a precise local or volume-averaged definition must be stated.
W	MHD energy functional	Total magnetic plus internal energy used in a variational equilibrium formulation.
γ_{ad}	Adiabatic index	Ratio of specific heats in an ideal-gas pressure model; not an instability growth rate.
ω	Angular frequency	Oscillation rate in radians per second.
$k_{\parallel}$	Parallel wave number	Spatial phase variation per unit length along $\mathbf{B}$.

1 The physical problem: confining a hot plasma

1.1 What is being confined?

A plasma is an ionized gas containing positively charged ions, negatively charged electrons, and sometimes neutral particles. In a fusion plasma the temperature is so high that ordinary material walls cannot directly contain the central plasma without being rapidly cooled or damaged. Magnetic confinement instead uses the Lorentz force

$$\mathbf{F}_s = q_s \left(\mathbf{E} + \mathbf{v}_s \times \mathbf{B}\right) \tag{1.1}$$

to influence charged-particle motion. Here $\mathbf{F}_s$ is the force on a particle of species s, q_s is its charge, $\mathbf{v}_s$ is its velocity, $\mathbf{E}$ is the electric field, and $\mathbf{B}$ is the magnetic flux density.

The magnetic part of the Lorentz force is perpendicular to the instantaneous particle velocity. It therefore bends a charged particle's path without directly changing its kinetic energy. In a sufficiently strong and slowly varying magnetic field, a particle executes rapid gyromotion around a magnetic field line while moving more freely along the field. This anisotropy is the basic reason magnetic fields can reduce transport across the field.

Central confinement idea. A strong magnetic field does not place particles at fixed points. It organizes their trajectories so that cross-field motion is much slower than motion along the field. Successful confinement therefore requires field lines themselves to remain inside the device for long distances and, ideally, to lie on closed toroidal surfaces.

1.2 Why close the magnetic field into a torus?

If a magnetic field line ends on a material wall, particles streaming along it can be lost rapidly. A straight magnetic bottle can reflect some particles using stronger magnetic fields at its ends, but end losses remain a major challenge. Toroidal devices bend the plasma and magnetic field into a ring so that field lines need not terminate at physical end plates.

A torus has two natural directions:

- the **toroidal direction**, which goes the long way around the ring;
- the **poloidal direction**, which goes the short way around a cross-section of the ring.

A useful confining field contains both toroidal and poloidal components. Their combination makes a field line wind helically around the torus instead of repeatedly sampling only one side of the device.

1.3 Tokamak and stellarator strategies

A **tokamak** produces most of its toroidal field using external coils and a substantial part of its poloidal field using a large toroidal plasma current. A **stellarator** produces the required three-dimensional field-line twist primarily through external coils and shaped magnetic geometry. The stellarator therefore does not require a large transformer-driven plasma current for its fundamental confinement geometry.

This difference has several consequences.

1. A stellarator equilibrium is intrinsically three-dimensional: magnetic quantities generally depend on both poloidal and toroidal angles.
2. The rotational transform can be sustained in steady state by external coils.
3. Coil and boundary shape become direct design variables for confinement, stability, and energetic-particle behavior.
4. Numerical representation is more demanding because axisymmetry cannot usually reduce the problem to two dimensions.

Equilibrium is not confinement quality. A magnetic configuration may satisfy force balance yet have poor particle confinement, undesirable neoclassical transport, large fast-ion losses, or unstable waves. Module 1 determines the equilibrium geometry. Modules 3–7 determine important consequences of that geometry.

2 From particles to magnetohydrodynamics

2.1 Why use a fluid model?

A kinetic description follows distribution functions in six-dimensional phase space: three position coordinates and three velocity coordinates. That description is essential for resonant energetic particles and is developed later in the project. The bulk thermal plasma, however, can often be represented on large spatial and temporal scales using fluid variables such as mass density $\rho_m(\mathbf{x}, t)$, flow velocity $\mathbf{u}(\mathbf{x}, t)$, and pressure $p(\mathbf{x}, t)$.

Magnetohydrodynamics, abbreviated MHD, combines fluid dynamics with Maxwell's equations. The ideal-MHD model assumes, among other approximations, high electrical conductivity, a single bulk fluid velocity, and usually an isotropic scalar pressure. These assumptions are not universally valid, but they provide the standard starting point for macroscopic equilibrium and stability; a systematic development is given by Freidberg [1].

2.2 Ideal-MHD equations before taking the equilibrium limit

A common ideal-MHD system is

$$\frac{\partial \rho_m}{\partial t} + \nabla \cdot (\rho_m \mathbf{u}) = 0, \tag{2.1}$$

$$\rho_m \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mathbf{J} \times \mathbf{B}, \tag{2.2}$$

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{u} \times \mathbf{B}), \tag{2.3}$$

$$\nabla \cdot \mathbf{B} = 0, \tag{2.4}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}, \tag{2.5}$$

$$\frac{d}{dt}\left(\frac{p}{\rho_m^{\gamma_{\mathrm{ad}}}}\right) = 0. \tag{2.6}$$

Equation (2.1) conserves mass. Equation (2.2) is Newton's second law for the fluid. Equation (2.3) evolves the magnetic field in an ideal conductor. Equation (2.4) states that magnetic monopoles are absent. Equation (2.5) is Ampère's law without the displacement-current term, appropriate when characteristic speeds are much smaller than the speed of light. Equation (2.6) is one possible pressure closure, with γ_{ad} the adiabatic index.

The material derivative is

$$\frac{d}{dt} = \frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla. \tag{2.7}$$

It measures the rate of change experienced by a fluid element moving with velocity $\mathbf{u}$.

2.3 Assumptions used for the baseline equilibrium

The baseline Module 1 equilibrium adopts the following conceptual model unless a future implementation explicitly states otherwise:

1. **Time independence:** $\partial/\partial t = 0$.
2. **No equilibrium bulk flow:** $\mathbf{u} = \mathbf{0}$.
3. **Scalar pressure:** the pressure is represented by one scalar field p, rather than separate parallel and perpendicular pressures.
4. **Ideal magnetostatic force balance:** inertia, viscosity, and resistive electric-field effects are neglected in the equilibrium equation.
5. **Nested surfaces for the primary model:** the confined region is represented by smooth, nonintersecting magnetic surfaces. Islands and stochastic regions are discussed as limitations rather than included in the baseline representation.

With these assumptions, Equation (2.2) reduces to the static equilibrium equation developed next.

3 Magnetostatic equilibrium equations

3.1 The three governing relations

The mathematical anchor for Module 1 is

$$\nabla p = \mathbf{J} \times \mathbf{B}, \tag{3.1}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}, \tag{3.2}$$

$$\nabla \cdot \mathbf{B} = 0. \tag{3.3}$$

Equation (3.1) balances the pressure-gradient force against the magnetic Lorentz force density. Equation (3.2) relates current density to magnetic-field curl. Equation (3.3) requires the magnetic field to be divergence free.

The units provide an immediate check. Pressure gradient has units

$$[\nabla p] = \mathrm{Pa\,m^{-1}} = \mathrm{N\,m^{-3}}. \tag{3.4}$$

The Lorentz force density has units

$$[\mathbf{J} \times \mathbf{B}] = \left(\mathrm{A\,m^{-2}}\right)(\mathrm{T}) = \mathrm{N\,m^{-3}}, \tag{3.5}$$

so the two sides of Equation (3.1) are dimensionally consistent.

3.2 Pressure is constant along a magnetic field line

Take the dot product of Equation (3.1) with $\mathbf{B}$:

$$\mathbf{B} \cdot \nabla p = \mathbf{B} \cdot (\mathbf{J} \times \mathbf{B}) = 0. \tag{3.6}$$

A scalar field changes along a field line at a rate proportional to $\mathbf{B} \cdot \nabla p$. Therefore Equation (3.6) states that pressure is constant along each magnetic field line.

If a field line densely covers a smooth magnetic surface, continuity implies that pressure is constant over the entire surface. We may then write

$$p = p(\psi), \tag{3.7}$$

where ψ labels magnetic surfaces. A quantity that depends only on the surface label is called a **flux function**.

Taking the dot product of Equation (3.1) with $\mathbf{J}$ similarly gives

$$\mathbf{J} \cdot \nabla p = 0. \tag{3.8}$$

Thus both magnetic field lines and current lines lie tangent to constant-pressure surfaces in this ideal nested-surface equilibrium.

3.3 Perpendicular and parallel current

Decompose current density into components perpendicular and parallel to **B**:

$$\mathbf{J} = \mathbf{J}_\perp + \mathbf{J}_\parallel, \qquad \mathbf{J}_\parallel = \frac{\mathbf{J}\cdot\mathbf{B}}{B^2}\mathbf{B}. \tag{3.9}$$

Cross Equation (3.1) with **B**:

$$\mathbf{B}\times\nabla p = \mathbf{B}\times(\mathbf{J}\times\mathbf{B}). \tag{3.10}$$

Using the vector identity

$$\mathbf{A}\times(\mathbf{C}\times\mathbf{D}) = \mathbf{C}(\mathbf{A}\cdot\mathbf{D}) - \mathbf{D}(\mathbf{A}\cdot\mathbf{C}), \tag{3.11}$$

we obtain

$$\mathbf{B}\times(\mathbf{J}\times\mathbf{B}) = B^2\mathbf{J} - (\mathbf{B}\cdot\mathbf{J})\mathbf{B} = B^2\mathbf{J}_\perp. \tag{3.12}$$

Therefore

$$\boxed{\mathbf{J}_\perp = \frac{\mathbf{B}\times\nabla p}{B^2}.} \tag{3.13}$$

The perpendicular current is directly fixed by pressure gradients and magnetic geometry. The parallel current is not determined by Equation (3.1) alone; it is constrained together with geometry by Ampère's law and charge conservation.

3.4 Magnetic pressure and magnetic tension

Substitute Equation (3.2) into the magnetic force density:

$$\mathbf{J}\times\mathbf{B} = \frac{1}{\mu_0}(\nabla\times\mathbf{B})\times\mathbf{B}. \tag{3.14}$$

Using $\nabla\cdot\mathbf{B} = 0$, this can be written as

$$\mathbf{J}\times\mathbf{B} = -\nabla\left(\frac{B^2}{2\mu_0}\right) + \frac{1}{\mu_0}(\mathbf{B}\cdot\nabla)\mathbf{B}. \tag{3.15}$$

The first term is the gradient of **magnetic pressure**, $B^2/(2\mu_0)$. The second term is **magnetic tension**, associated with bending field lines.

Writing $\mathbf{B} = B\mathbf{b}$ gives

$$(\mathbf{B}\cdot\nabla)\mathbf{B} = B(\mathbf{b}\cdot\nabla B)\mathbf{b} + B^2\boldsymbol{\kappa}, \tag{3.16}$$

where

$$\boldsymbol{\kappa} = \mathbf{b}\cdot\nabla\mathbf{b} \tag{3.17}$$

is the magnetic curvature vector. Perpendicular force balance can consequently be interpreted as a competition among plasma pressure, magnetic-pressure variation, and field-line tension.

Physical picture of equilibrium. Plasma pressure tends to expand the plasma. Magnetic-pressure gradients and the tension of curved field lines provide restoring forces. A valid equilibrium is a three-dimensional arrangement in which these forces cancel at every point inside the plasma.

3.5 Plasma beta

A local pressure-to-magnetic-pressure ratio is

$$\beta(\mathbf{x}) = \frac{p(\mathbf{x})}{B^2(\mathbf{x})/(2\mu_0)} = \frac{2\mu_0 p(\mathbf{x})}{B^2(\mathbf{x})}. \tag{3.18}$$

A volume-averaged beta may be defined by

$$\langle\beta\rangle_V = \frac{2\mu_0 \int_V p\, dV}{\int_V B^2\, dV}. \tag{3.19}$$

Different communities use slightly different averages, so a reported beta value must include its definition. Increasing beta strengthens the influence of plasma pressure on the magnetic geometry and can shift the magnetic axis, rotational transform, and stability properties.

3.6 Equilibrium does not guarantee stability

Equation (3.1) states only that the net force vanishes for the chosen state. It does not state that a displaced plasma element experiences a restoring force. Stability requires perturbing the equilibrium and determining whether perturbations oscillate, decay, or grow. Module 4 begins that process for reduced shear-Alfvén waves. In symbols,

$$\text{equilibrium: } \mathbf{F}(\mathbf{U}_0) = 0, \qquad \text{stability: behavior of } \delta\mathbf{U} \text{ near } \mathbf{U}_0. \tag{3.20}$$

Here $\mathbf{U}_0$ denotes the equilibrium state and $\delta\mathbf{U}$ denotes a small perturbation.

4 Coordinate systems for a toroidal plasma

4.1 Cartesian and cylindrical coordinates

Cartesian coordinates (x, y, z) use fixed orthonormal basis vectors. For a toroidal device, cylindrical coordinates (R, ϕ, Z) are more natural:

$$x = R\cos\phi, \qquad y = R\sin\phi, \qquad z = Z. \tag{4.1}$$

The coordinate $R \geq 0$ is distance from the vertical symmetry axis, ϕ is the geometric toroidal angle, and Z is vertical position.

The cylindrical basis vectors $\mathbf{e}_R$, $\mathbf{e}_\phi$, and $\mathbf{e}_Z$ vary with ϕ. A vector field may be written

$$\mathbf{B} = B_R \mathbf{e}_R + B_\phi \mathbf{e}_\phi + B_Z \mathbf{e}_Z. \tag{4.2}$$

In an axisymmetric device, scalar quantities are independent of ϕ. In a stellarator, this simplification generally does not hold.

4.2 Major radius, minor radius, and aspect ratio

The **magnetic axis** is the central closed field line around which nested magnetic surfaces are organized. A characteristic major radius R_0 measures the distance from the device's central axis to the magnetic axis. A characteristic minor radius a measures the radial size of the plasma cross-section. The aspect ratio is

$$A = \frac{R_0}{a}. \tag{4.3}$$

Large aspect ratio means the torus is relatively thin compared with its major radius. Reduced analytic models often exploit $a/R_0 \ll 1$, but realistic stellarator equilibria need not be accurately represented by this limit.

4.3 Poloidal and toroidal angles

A geometric poloidal angle may be introduced around a chosen cross-sectional center, while the geometric toroidal angle is ϕ. Magnetic calculations often use modified angles (θ, ζ) chosen to simplify field-line equations rather than to match geometric angles exactly. It is therefore essential to state whether an angle is geometric, computational, or a straight-field-line magnetic angle.

Both angular coordinates are periodic:

$$\theta \equiv \theta + 2\pi, \qquad \zeta \equiv \zeta + 2\pi. \tag{4.4}$$

If the device has N_{fp} identical field periods, physical geometry repeats after

$$\Delta\zeta = \frac{2\pi}{N_{\text{fp}}}. \tag{4.5}$$

5 Magnetic field lines and flux surfaces

5.1 Definition of a magnetic field line

A magnetic field line is a curve $\mathbf{x}(\ell)$ whose tangent is parallel to $\mathbf{B}$. One convenient parameter is arc length ℓ, for which

$$\frac{d\mathbf{x}}{d\ell} = \mathbf{b}(\mathbf{x}) = \frac{\mathbf{B}(\mathbf{x})}{B(\mathbf{x})}. \tag{5.1}$$

An alternative parameter τ may be used:

$$\frac{d\mathbf{x}}{d\tau} = \mathbf{B}(\mathbf{x}). \tag{5.2}$$

Changing the parameter changes how quickly the curve is traversed, not the geometric path.

Field lines are mathematical integral curves. They are not material strings, although the magnetic-tension analogy is often useful. In ideal MHD, flux freezing gives them a strong dynamical connection to the plasma motion.

5.2 Definition of a magnetic surface

A magnetic surface is a surface labeled by a scalar $\psi(\mathbf{x})$ such that

$$\mathbf{B} \cdot \nabla\psi = 0. \tag{5.3}$$

The gradient $\nabla\psi$ is normal to a constant-ψ surface. Equation (5.3) therefore says that $\mathbf{B}$ is tangent to the surface at every point. A field line that begins on the surface remains on it.

For nested surfaces, the plasma region resembles a collection of nonintersecting toroidal shells:

$$0 \leq s \leq 1, \tag{5.4}$$

where $s = 0$ labels the magnetic axis and $s = 1$ labels the chosen plasma boundary, commonly the last closed flux surface.

5.3 Toroidal and poloidal magnetic flux

Magnetic flux through an oriented surface S is

$$\Psi = \int_S \mathbf{B} \cdot d\mathbf{S}, \tag{5.5}$$

where $d\mathbf{S} = \mathbf{n}\, dS$ and $\mathbf{n}$ is the surface normal.

For a toroidal magnetic surface, two fluxes are useful:

- The **toroidal flux** Ψ_t passes through a poloidal cross-section of the torus.

- The **poloidal flux** Ψ_p passes through a toroidal ribbon extending around the long direction.

Sign conventions vary. This project will always store the convention explicitly rather than inferring it from a variable name.

A common normalized surface coordinate is

$$s = \frac{\Psi_t}{\Psi_{t,a}}, \tag{5.6}$$

where $\Psi_{t,a}$ is the toroidal flux at the selected plasma boundary. Another common radial coordinate is

$$\rho = \sqrt{s}. \tag{5.7}$$

For a simple circular plasma with nearly uniform toroidal field, toroidal flux scales approximately with the square of minor radius, so ρ behaves approximately like a normalized geometric radius. In shaped three-dimensional geometry, ρ remains a flux label rather than a literal Euclidean distance.

5.4 Closed, islanded, and stochastic field-line regions

The nested-surface picture is powerful but not universal.

- On an **irrational surface**, a field line generally does not close after a finite number of turns and can densely cover the surface.
- On a **rational surface**, the field-line winding ratio is rational; perturbations can generate magnetic islands.
- In a **stochastic region**, neighboring field lines separate chaotically and a single smooth global surface label may not exist.

A nested-surface equilibrium code such as standard VMEC represents the first category and idealized rational surfaces through smooth surfaces, but does not directly resolve magnetic islands or stochastic volumes. Other equilibrium formulations are needed when those structures are essential.

6 Rotational transform, safety factor, and magnetic shear

6.1 Rotational transform

Follow a field line and count how its poloidal angle changes as the toroidal angle increases. In straight-field-line coordinates, the rotational transform is

$$\iota(s) = \frac{d\theta}{d\zeta} \tag{6.1}$$

along a field line, with the derivative understood as an average winding rate on the surface. Equivalently,

$$\iota(s) = \lim_{N\to\infty} \frac{\Delta\theta}{\Delta\zeta} \tag{6.2}$$

for a trajectory followed through many toroidal turns.

If $\iota = 0.4$, the field line advances on average 0.4 poloidal turns per toroidal turn. After five toroidal turns it advances two poloidal turns. If $\iota = 2/5$ exactly, the field line is rational in the idealized straight-field-line representation and closes after a finite number of turns.

6.2 Safety factor

Tokamak literature often uses the safety factor

$$q(s) = \frac{1}{\iota(s)}. \tag{6.3}$$

With this convention, q counts toroidal turns per poloidal turn. Some sign conventions include a minus sign depending on angle orientation and magnetic-field direction. This project uses Equation (6.3) unless a data source documents a different convention.

6.3 Rational surfaces

A rational surface satisfies

$$\iota(s_r) = \frac{n}{m}, \tag{6.4}$$

where m and n are nonzero integers with no common factor and s_r is the rational surface location. A field line then closes after m toroidal-angle periods and n poloidal-angle periods under the stated convention. The same integers appear in Fourier perturbations, resonances, and Alfvén-continuum conditions.

6.4 Magnetic shear

Magnetic shear describes radial change in field-line pitch. Several definitions are used. A common dimensionless form in terms of ι is

$$\hat{s}_\iota(s) = \frac{s}{\iota(s)} \frac{d\iota}{ds}. \tag{6.5}$$

In terms of safety factor,

$$\hat{s}_q(s) = \frac{s}{q(s)} \frac{dq}{ds} = -\hat{s}_\iota(s). \tag{6.6}$$

The sign difference follows from $q = 1/\iota$. One must therefore report the definition, not only the phrase "magnetic shear."

Shear matters because nearby field lines accumulate different angular phases. It affects magnetic-island overlap, wave continua, resonant surfaces, and radial mode structure.

6.5 Connection to the Module 4 parallel wave number

Consider a perturbation with phase

$$S = m\theta - n\zeta - \omega t, \tag{6.7}$$

where m is the poloidal mode number, n is the toroidal mode number, and ω is angular frequency. Along a field line,

$$\frac{dS}{d\zeta} = m\frac{d\theta}{d\zeta} - n = m\iota - n. \tag{6.8}$$

The parallel wave number is the phase gradient along the field:

$$k_\parallel = \mathbf{b} \cdot \nabla S. \tag{6.9}$$

In a simple large-aspect-ratio approximation, $\mathbf{b} \cdot \nabla\zeta \approx 1/R_0$, so

$$k_{\parallel,mn}(s) \approx \frac{m\iota(s) - n}{R_0}. \tag{6.10}$$

Module 4 may use the algebraically equivalent opposite sign $(n - m\iota)/R_0$ because its eigenfrequency depends on $k_\parallel^2$. The sign becomes important when propagation direction and resonant particle motion are retained.

Why Module 1 controls Module 4. The equilibrium supplies $B(s, \theta, \zeta)$, mass density, metric information, and $\iota(s)$. These determine the local Alfvén speed and the parallel wavelength of each harmonic. Consequently, changing the equilibrium changes the Alfvén continuum and the locations at which energetic particles can resonate with waves.

7 Magnetic flux coordinates

7.1 Coordinate mapping

Let

$$(u^1, u^2, u^3) = (s, \theta, \zeta) \tag{7.1}$$

be curvilinear coordinates. The physical position is a mapping

$$\mathbf{x} = \mathbf{x}(s, \theta, \zeta). \tag{7.2}$$

The covariant basis vectors are

$$\mathbf{e}_i = \frac{\partial \mathbf{x}}{\partial u^i}, \tag{7.3}$$

and the reciprocal, or contravariant, basis vectors are

$$\mathbf{e}^i = \nabla u^i. \tag{7.4}$$

They satisfy

$$\mathbf{e}_i \cdot \mathbf{e}^j = \delta_i{}^j, \tag{7.5}$$

where $\delta_i{}^j$ is the Kronecker delta, equal to one when $i = j$ and zero otherwise.

7.2 Metric tensors

The covariant metric tensor is

$$g_{ij} = \mathbf{e}_i \cdot \mathbf{e}_j, \tag{7.6}$$

and the contravariant metric tensor is

$$g^{ij} = \mathbf{e}^i \cdot \mathbf{e}^j. \tag{7.7}$$

The matrices $[g_{ij}]$ and $[g^{ij}]$ are inverses. They convert coordinate derivatives and components into physical lengths and dot products.

For example, the squared differential distance is

$$d\ell^2 = g_{ij}\, du^i du^j, \tag{7.8}$$

where repeated indices are summed from 1 to 3.

7.3 Jacobian and volume element

The coordinate Jacobian is

$$\sqrt{g} = \mathbf{e}_s \cdot (\mathbf{e}_\theta \times \mathbf{e}_\zeta). \tag{7.9}$$

The physical volume element is

$$dV = \sqrt{g}\, ds\, d\theta\, d\zeta. \tag{7.10}$$

The sign of $\sqrt{g}$ depends on coordinate orientation. A production interface should either enforce a positive orientation or store the sign convention explicitly.

7.4 Contravariant representation of a tangent, divergence-free field

A particularly useful straight-field-line representation is

$$\boxed{\mathbf{B} = \nabla\psi \times \nabla\theta + \iota(\psi)\,\nabla\zeta \times \nabla\psi.} \tag{7.11}$$

Here ψ is a toroidal-flux-like surface coordinate. Equation (7.11) automatically gives

$$\mathbf{B} \cdot \nabla\psi = 0 \tag{7.12}$$

and, provided ι depends only on ψ, satisfies $\nabla \cdot \mathbf{B} = 0$.

Using the Jacobian identity

$$(\nabla\psi \times \nabla\theta) \cdot \nabla\zeta = \frac{1}{\sqrt{g}}, \tag{7.13}$$

we find

$$\mathbf{B} \cdot \nabla\zeta = \frac{1}{\sqrt{g}}, \tag{7.14}$$

$$\mathbf{B} \cdot \nabla\theta = \frac{\iota}{\sqrt{g}}. \tag{7.15}$$

Therefore

$$\frac{d\theta}{d\zeta} = \frac{\mathbf{B} \cdot \nabla\theta}{\mathbf{B} \cdot \nabla\zeta} = \iota, \tag{7.16}$$

which explains the phrase **straight-field-line coordinates**: in the (θ, ζ) plane, field lines are straight lines of slope ι.

7.5 Covariant representation and Boozer coordinates

Any vector can also be expanded in gradients:

$$\mathbf{B} = B_s\nabla s + B_\theta\nabla\theta + B_\zeta\nabla\zeta. \tag{7.17}$$

In Boozer coordinates the covariant representation can be written

$$\mathbf{B} = K(s,\theta,\zeta)\nabla s + I(s)\nabla\theta + G(s)\nabla\zeta. \tag{7.18}$$

The functions $I(s)$ and $G(s)$ are flux functions related to net toroidal and poloidal currents, while K may vary over a surface. Exact multiplicative and sign conventions depend on how flux and angles are normalized. This is why a geometry file must store metadata rather than only arrays.

Boozer coordinates are useful because the magnetic-field-line trajectories are straight and the covariant angular components are flux functions. Guiding-center orbit equations and quasisymmetry are especially transparent in this representation. Detailed treatments of flux coordinates and the original Boozer construction are given in Refs. [2, 6].

7.6 Hamada coordinates

Hamada coordinates are another straight-field-line system. They are constructed so that both magnetic field lines and current lines have simple contravariant representations and the Jacobian is a flux function under suitable conditions. Boozer and Hamada coordinates serve different analytical and numerical purposes; neither is simply a geometric angle system.

7.7 Coordinate singularity at the magnetic axis

At $s = 0$, all poloidal angles refer to the same physical magnetic axis. The coordinate system is therefore singular in the same way that polar angle is undefined at the origin. This is a coordinate singularity, not necessarily a physical singularity. Numerical representations must impose regularity conditions so that physical fields remain finite and single-valued near the axis.

8 Flux-surface averages and radial geometry

8.1 Surface average

For a scalar field $A(s, \theta, \zeta)$, a common flux-surface average is

$$\langle A \rangle_s = \frac{\displaystyle\int_0^{2\pi} \int_0^{2\pi} A\sqrt{g}\, d\theta\, d\zeta}{\displaystyle\int_0^{2\pi} \int_0^{2\pi} \sqrt{g}\, d\theta\, d\zeta}. \tag{8.1}$$

The Jacobian appears because equal increments of (θ, ζ) do not generally correspond to equal physical surface or volume weighting.

8.2 Enclosed volume and differential volume

The volume inside surface s is

$$V(s) = \int_0^s ds' \int_0^{2\pi} d\theta \int_0^{2\pi} d\zeta\, \sqrt{g}(s', \theta, \zeta). \tag{8.2}$$

Its derivative is

$$V'(s) = \frac{dV}{ds} = \int_0^{2\pi} \int_0^{2\pi} \sqrt{g}\, d\theta\, d\zeta. \tag{8.3}$$

The function $V'(s)$ appears in one-dimensional transport equations. For example, a conservative radial balance often contains

$$\frac{1}{V'}\frac{\partial}{\partial s}\left(V'\Gamma_s\right), \tag{8.4}$$

where Γ_s is a radial flux measured per appropriate surface area and coordinate normalization.

8.3 Effective minor radius

A coordinate-independent size measure may be defined from volume:

$$a_{\text{eff}}(s) = \sqrt{\frac{V(s)}{2\pi^2 R_{\text{ref}}}}, \tag{8.5}$$

where R_{ref} is a stated reference major radius. This definition is motivated by the volume $2\pi^2 R_0 a^2$ of a circular torus. Other effective-radius definitions exist, so the formula must accompany the value.

9 Three-dimensional stellarator surface geometry

9.1 Why use Fourier series?

The angles θ and ζ are periodic. Any sufficiently smooth periodic surface shape can therefore be expanded in Fourier modes. A common representation in cylindrical coordinates is

$$R(s,\theta,\zeta) = \sum_{m,n} R_{mn}(s)\cos\left(m\theta - nN_{\text{fp}}\zeta\right), \tag{9.1}$$

$$Z(s,\theta,\zeta) = \sum_{m,n} Z_{mn}(s)\sin\left(m\theta - nN_{\text{fp}}\zeta\right). \tag{9.2}$$

Here m is a poloidal integer, n is a toroidal Fourier integer per field period, and N_{fp} is the number of field periods. The coefficients $R_{mn}(s)$ and $Z_{mn}(s)$ determine the shape of every flux surface.

The physical position is then

$$\mathbf{x}(s,\theta,\zeta) = R\cos\zeta\,\mathbf{e}_x + R\sin\zeta\,\mathbf{e}_y + Z\,\mathbf{e}_z, \tag{9.3}$$

when ζ is chosen as the geometric toroidal angle. If ζ is a magnetic toroidal angle, an additional mapping to the geometric angle is required.

9.2 Meaning of selected Fourier modes

The coefficient R_{00} largely sets a major-radius offset. The $m = 1, n = 0$ terms describe a basic circular or elliptical cross-sectional displacement. Higher poloidal mode numbers describe triangularity, squareness, and finer shaping. Nonzero toroidal mode numbers introduce variation from one toroidal location to another and are essential to a stellarator.

A Fourier coefficient should not be interpreted in isolation without knowing the angular convention. The same physical surface can have different coefficients after a change of poloidal angle.

9.3 Stellarator symmetry

A common discrete symmetry is stellarator symmetry. Under an appropriate choice of angular origins, the configuration is invariant under

$$(\theta, \zeta) \to (-\theta, -\zeta) \tag{9.4}$$

combined with a reflection in physical space. This symmetry motivates cosine terms for R and sine terms for Z in Equations (9.1)–(9.2). Configurations without stellarator symmetry require both sine and cosine coefficients in both coordinates.

9.4 Magnetic-field-strength spectrum

The magnetic-field magnitude can also be expanded:

$$B(s, \theta, \zeta) = \sum_{m,n} B_{mn}(s) \cos\left(m\theta - nN_{\mathrm{fp}}\zeta + \delta_{mn}(s)\right), \tag{9.5}$$

where δ_{mn} is a phase. With suitable symmetry and angle conventions, the phases can be absorbed into sine/cosine parity choices.

The surface shape spectrum and magnetic-strength spectrum are related through Maxwell's equations and equilibrium, but they are not identical. Particle trapping depends strongly on variation of B along a field line, so the B_{mn} spectrum is a critical Module 3 input.

9.5 Spectral resolution and aliasing

A numerical representation retains finite ranges

$$0 \le m \le m_{\mathrm{max}}, \qquad -n_{\mathrm{max}} \le n \le n_{\mathrm{max}}. \tag{9.6}$$

Increasing m_{max} and n_{max} resolves finer structure but increases computational cost and can amplify sensitivity to noisy input. Nonlinear products generate sums of mode numbers, so evaluation on an insufficient angular grid can cause aliasing: unresolved high-frequency content appears incorrectly as lower-frequency content. Production workflows should document spectral truncation and quadrature resolution.

10 Vacuum magnetic field and external coils

10.1 Biot–Savart law for a filamentary coil

A current-carrying wire contributes a magnetic field

$$\mathbf{B}(\mathbf{x}) = \frac{\mu_0 I_c}{4\pi} \oint_C \frac{d\boldsymbol{\ell}' \times (\mathbf{x} - \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|^3}, \tag{10.1}$$

where I_c is coil current, C is the coil centerline, $\mathbf{x}'$ is a source point on the coil, and $d\boldsymbol{\ell}'$ is an oriented line element. Fields from multiple coils superpose linearly in vacuum.

Real coils have finite cross-section, support structures, current-density variation, and manufacturing tolerances. A filament model is often adequate for conceptual studies but must be refined for engineering analysis near the conductor.

10.2 Vacuum equations

In a current-free vacuum region,

$$\nabla \times \mathbf{B} = 0, \qquad \nabla \cdot \mathbf{B} = 0. \tag{10.2}$$

Locally one may introduce a magnetic scalar potential Φ_m such that

$$\mathbf{B} = -\nabla \Phi_m, \qquad \nabla^2 \Phi_m = 0, \tag{10.3}$$

provided the region is simply connected or global topological fluxes are treated separately. Alternatively, a vector potential $\mathbf{A}$ may be used:

$$\mathbf{B} = \nabla \times \mathbf{A}, \tag{10.4}$$

which automatically enforces $\nabla \cdot \mathbf{B} = 0$.

10.3 Coil field versus plasma response

The total field inside an operating plasma is

$$\mathbf{B}_{\text{total}} = \mathbf{B}_{\text{coils}} + \mathbf{B}_{\text{plasma}}, \tag{10.5}$$

where $\mathbf{B}_{\text{plasma}}$ is generated by equilibrium plasma currents. At low beta and low net plasma current, the coil field may dominate. At finite beta, diamagnetic and parallel currents modify the geometry. A vacuum field-line calculation is therefore not automatically a finite-pressure MHD equilibrium.

11 Boundary conditions and equilibrium classes

11.1 Fixed-boundary equilibrium

In a fixed-boundary calculation, the plasma boundary shape is prescribed:

$$\mathbf{x}_a(\theta, \zeta) = \mathbf{x}(s = 1, \theta, \zeta). \tag{11.1}$$

The boundary is required to be a magnetic surface,

$$\mathbf{B} \cdot \mathbf{n} = 0 \qquad \text{on } s = 1, \tag{11.2}$$

where $\mathbf{n}$ is the outward unit normal. The solver determines the interior surfaces and field consistent with the pressure and current profiles.

Fixed-boundary calculations are useful when a target plasma shape is already known or when equilibrium is embedded inside an optimization loop that treats boundary shape as the design variable.

11.2 Free-boundary equilibrium

In a free-boundary calculation, external coils and currents are prescribed, and the plasma boundary is solved as part of the coupled plasma-vacuum problem. The interface must satisfy normal-field and stress-balance conditions. In the idealized case of a tangential magnetic field at the boundary, total pressure balance is

$$\left[p + \frac{B^2}{2\mu_0}\right]_{\text{inside}} = \left[\frac{B^2}{2\mu_0}\right]_{\text{outside}}, \tag{11.3}$$

where the outside is vacuum and square brackets here indicate evaluation on the two sides of the interface. Surface currents may permit a jump in tangential magnetic field, so a complete formulation must state whether such currents are allowed.

11.3 Magnetic axis regularity

At the magnetic axis, physical quantities must remain finite and single-valued. Fourier components must scale with radius in a manner consistent with regularity. For example, non-axisymmetric poloidal harmonics should vanish with appropriate powers of the minor-radius coordinate as $s \to 0$. Numerical solvers enforce this through basis choices, parity conditions, or explicit constraints.

11.4 Profile boundary conditions

Pressure typically decreases from a central value to a small edge value:

$$p(0) = p_0, \qquad p(1) \approx 0. \tag{11.4}$$

The exact edge condition is a model choice. A current profile may be specified through toroidal current, parallel current, or derivatives of flux functions. Profiles must be differentiable enough for the chosen numerical method; sharp edge gradients require adequate resolution.

12 Pressure, density, temperature, and current profiles

12.1 Pressure profile

A simple analytic pressure profile is

$$p(s) = p_0(1-s)^{\alpha_p}, \tag{12.1}$$

where p_0 is central pressure and $\alpha_p > 0$ controls peaking. Its derivative is

$$\frac{dp}{ds} = -\alpha_p p_0 (1-s)^{\alpha_p - 1}. \tag{12.2}$$

The pressure gradient, not only the pressure value, drives perpendicular current through Equation (3.13).

For multiple species with isotropic temperatures,

$$p(s) = \sum_s n_s(s) k_{\mathrm{B}} T_s(s), \tag{12.3}$$

where k_{B} is Boltzmann's constant. If temperature is specified in electronvolts, the conversion to joules must be handled consistently.

12.2 Mass density profile

The MHD mass density is

$$\rho_m(s) = \sum_s m_s n_s(s). \tag{12.4}$$

Electron mass is often negligible in the mass sum but not in charge balance or kinetic physics. Module 4 uses ρ_m to compute the Alfvén speed

$$v_A = \frac{B}{\sqrt{\mu_0 \rho_m}}. \tag{12.5}$$

Thus a geometry product that omits density is insufficient for a fully physical continuum calculation, even though density may be supplied by Module 7 rather than solved by Module 1.

12.3 Current-profile measures

In a stellarator, several current measures can be relevant:

- net toroidal plasma current;
- bootstrap current driven by collisional transport in pressure gradients;
- externally driven current from neutral beams or radio-frequency waves;
- Pfirsch–Schluter currents required to maintain current continuity in three-dimensional geometry;
- diamagnetic current associated with pressure gradients.

These categories are not independent arbitrary additions; they are different physical contributions to the total current density that must remain consistent with force balance and Ampere's law.

12.4 Profile consistency

An equilibrium solver cannot accept an arbitrary collection of mutually incompatible functions. Pressure, current, toroidal flux, and rotational transform are linked by the equilibrium equations. Depending on the solver formulation, one specifies some profiles and solves for the others. The input contract must state which quantities are prescribed and which are outputs.

Common modeling error. Supplying a desired pressure profile, a desired rotational-transform profile, and a desired current profile does not guarantee that all three can coexist with a prescribed boundary. An equilibrium code enforces compatibility; it is not merely an interpolation tool.

13 Axisymmetry versus fully three-dimensional equilibrium

13.1 Axisymmetric reduction

An axisymmetric equilibrium is invariant under toroidal rotation:

$$\frac{\partial}{\partial \phi} = 0. \tag{13.1}$$

This continuous symmetry produces a conserved canonical toroidal momentum for particle motion and permits the magnetic field to be represented using a two-dimensional poloidal-flux function. The equilibrium equations reduce to the Grad–Shafranov equation, a nonlinear elliptic partial differential equation in the (R, Z) plane.

The exact Grad–Shafranov form depends on flux normalization. A common form is

$$R\frac{\partial}{\partial R}\left(\frac{1}{R}\frac{\partial\psi_p}{\partial R}\right) + \frac{\partial^2\psi_p}{\partial Z^2} = -\mu_0 R^2\frac{dp}{d\psi_p} - F\frac{dF}{d\psi_p}, \tag{13.2}$$

where $\psi_p(R, Z)$ is a poloidal-flux function and $F(\psi_p) = RB_\phi$. This equation is included to show what axisymmetry buys mathematically: a scalar two-dimensional equilibrium equation.

13.2 Why a stellarator is harder

A stellarator generally has only discrete toroidal periodicity, not continuous axisymmetry. Quantities depend on all three coordinates:

$$A = A(s, \theta, \zeta). \tag{13.3}$$

There is no single two-dimensional Grad–Shafranov equation that describes a general three-dimensional stellarator equilibrium. Instead, numerical methods solve coupled equations for the shapes of many nested surfaces and for field/current quantities over each surface.

Three-dimensionality affects more than computational cost. It changes the confinement physics summarized in the review by Helander [3]:

- magnetic-well structure and trapped-particle orbits;
- bootstrap and Pfirsch–Schluter currents;
- neoclassical transport;
- energetic-particle confinement;
- coupling among poloidal and toroidal wave harmonics;
- coil complexity and sensitivity to errors.

14 Quasisymmetry and related optimization ideas

14.1 Why symmetry matters for particle confinement

In an exactly axisymmetric magnetic field, the particle Lagrangian is independent of the toroidal angle. Noether's theorem then gives conservation of canonical toroidal momentum. This extra invariant strongly restricts radial particle excursions.

A stellarator lacks exact axisymmetry in physical space, but it can be designed so that the magnetic-field magnitude B has an approximate continuous symmetry in Boozer coordinates. This property is called **quasisymmetry**. The vector field remains three-dimensional, while the magnitude sampled by guiding centers has a simpler dependence.

14.2 General quasisymmetric form

A quasisymmetric magnetic-field strength has the form

$$B = B\left(s, M\theta - N\zeta\right), \tag{14.1}$$

where M and N are fixed integers that define the symmetry direction. The most common cases are:

- **quasi-axisymmetry:** $N = 0$, so B is approximately independent of the Boozer toroidal angle;
- **quasi-helical symmetry:** both M and N are nonzero, so B approximately follows a helical angle.

The word "quasi" is essential: exact global quasisymmetry is severely constrained, and practical configurations minimize symmetry-breaking Fourier components rather than eliminating all of them. Modern high-precision constructions are discussed by Landreman and Paul [9].

14.3 Quasi-isodynamic configurations

Another stellarator optimization strategy is quasi-isodynamicity, associated with favorable contours of the second adiabatic invariant and reduced radial drift of trapped particles. It is not the same condition as quasisymmetry. Both strategies aim to improve confinement but organize the magnetic geometry differently.

14.4 Why Module 1 should not reduce geometry to one score

A single quasisymmetry-error metric cannot fully characterize an equilibrium. A useful geometry database should retain the field spectrum, rotational transform, magnetic shear, curvature, magnetic wells, effective ripple measures, and coil/plasma constraints. Different downstream modules are sensitive to different features.

15 Geometric quantities needed for particle orbits

15.1 Field magnitude and unit vector

Guiding-center motion depends on

$$B = |\mathbf{B}|, \qquad \mathbf{b} = \frac{\mathbf{B}}{B}. \tag{15.1}$$

These must be available as continuous functions or sufficiently accurate interpolants in physical space or magnetic coordinates.

15.2 Magnetic-field gradient

The gradient

$$\nabla B \tag{15.2}$$

drives the grad-B drift. A noisy numerical derivative can cause artificial orbit diffusion even when B itself is accurate. Geometry export should therefore include either analytically differentiable spectral representations or derivative fields computed with controlled accuracy.

15.3 Field-line curvature

The curvature

$$\boldsymbol{\kappa} = \mathbf{b} \cdot \nabla \mathbf{b} \tag{15.3}$$

drives curvature drift and contributes to MHD interchange physics. Its dimensions are inverse length. The curvature vector is perpendicular to $\mathbf{b}$ because differentiating $\mathbf{b} \cdot \mathbf{b} = 1$ along the field gives

$$\mathbf{b} \cdot \boldsymbol{\kappa} = 0. \tag{15.4}$$

15.4 Mirror ratio and magnetic wells

On a chosen field line or surface, define

$$B_{\min}(s) = \min_{\theta,\zeta} B(s, \theta, \zeta), \qquad B_{\max}(s) = \max_{\theta,\zeta} B(s, \theta, \zeta). \tag{15.5}$$

The mirror ratio is

$$\mathcal{R}_m(s) = \frac{B_{\max}(s)}{B_{\min}(s)}. \tag{15.6}$$

Particles with sufficiently large perpendicular velocity can be reflected from high-B regions and become trapped. The detailed arrangement of local minima and maxima along a field line matters more than the mirror ratio alone.

15.5 Electric potential

Module 1 primarily supplies magnetic equilibrium, but guiding-center orbits may also require an electrostatic potential

$$\Phi = \Phi(s, \theta, \zeta), \qquad \mathbf{E} = -\nabla\Phi \tag{15.7}$$

for a static electrostatic field. A lowest-order model often assumes $\Phi = \Phi(s)$, giving a radial electric field. Because this potential is not determined by ideal magnetostatic force balance alone, its source and coupling location must be documented separately.

16 Geometric quantities needed for waves and stability

16.1 Parallel derivative

The derivative along the magnetic field is

$$\nabla_{\|} A = \mathbf{b} \cdot \nabla A \tag{16.1}$$

for a scalar A. In magnetic coordinates, the contravariant field representation provides

$$\mathbf{B} \cdot \nabla A = \frac{1}{\sqrt{g}} \left(\frac{\partial A}{\partial \zeta} + \iota \frac{\partial A}{\partial \theta} \right) \tag{16.2}$$

under the convention of Equation (7.11). This operator is central to shear-Alfven propagation.

16.2 Local Alfven frequency

For a Fourier harmonic proportional to $\exp[i(m\theta - n\zeta - \omega t)]$, a reduced local shear-Alfven frequency is

$$\omega_A(s, \theta, \zeta) \sim |k_{\|}| v_A, \qquad v_A = \frac{B}{\sqrt{\mu_0 \rho_m}}. \tag{16.3}$$

In a three-dimensional equilibrium, B, the metric, and sometimes density vary over a surface, so a full continuum calculation is more involved than the simple radial expression used in the reduced Module 4 benchmark.

16.3 Curvature and pressure-gradient coupling

MHD stability operators contain combinations of pressure gradient, current, field-line curvature, and metric coefficients. Bad curvature means that pressure forces tend to displace plasma farther in the direction of the original displacement; good curvature tends to provide restoration. In a three-dimensional torus, curvature varies strongly along a field line, so stability depends on weighted averages and mode structure rather than a single local sign.

16.4 Geodesic curvature

Curvature can be decomposed relative to a flux surface into normal and tangential components. Geodesic curvature lies within the surface and is perpendicular to the field line. It participates in coupling between shear-Alfven and compressional or acoustic responses. A production equilibrium interface should provide enough metric and derivative information to calculate this decomposition consistently.

17 A reduced analytic equilibrium for intuition

17.1 Circular nested surfaces

Before using a three-dimensional equilibrium, it is useful to define a manufactured geometry with circular concentric surfaces:

$$R(r,\theta) = R_0 + r\cos\theta, \tag{17.1}$$

$$Z(r,\theta) = r\sin\theta, \tag{17.2}$$

where $0 \le r \le a$. Let

$$s = \left(\frac{r}{a}\right)^2, \qquad r = a\sqrt{s}. \tag{17.3}$$

This model is axisymmetric and therefore not a stellarator, but it provides an interpretable verification limit for coordinate mappings, Jacobians, volume, and interpolation.

The enclosed volume is approximately

$$V(r) = 2\pi^2 R_0 r^2, \tag{17.4}$$

and hence

$$V(s) = 2\pi^2 R_0 a^2 s, \qquad V'(s) = 2\pi^2 R_0 a^2. \tag{17.5}$$

These relations give exact or near-exact checks for a large-aspect-ratio circular implementation.

17.2 Prescribed rotational transform

A simple transform profile is

$$\iota(s) = \iota_0 + \iota_1 s, \tag{17.6}$$

where ι_0 is the on-axis transform and ι_1 is the total edge-minus-axis change. The shear measure becomes

$$\hat{s}_\iota(s) = \frac{s\iota_1}{\iota_0 + \iota_1 s}. \tag{17.7}$$

This profile is not derived from force balance; it is a manufactured geometry input suitable for checking downstream equations.

17.3 Simple helical field-line tracing

In straight-field-line coordinates,

$$\frac{d\theta}{d\zeta} = \iota(s). \tag{17.8}$$

For constant s and constant ι,

$$\theta(\zeta) = \theta_0 + \iota\zeta, \tag{17.9}$$

where θ_0 is the initial poloidal angle. This exact solution is a basic field-line tracer test.

17.4 What the analytic model omits

The circular manufactured model omits three-dimensional shaping, finite-beta equilibrium shifts, realistic current distributions, magnetic wells, coil ripple, islands, and stochasticity. It is a verification case, not a predictive stellarator equilibrium.

18 Numerical solution of three-dimensional equilibrium

18.1 Unknowns and input choices

A nested-surface equilibrium calculation typically represents the position of each surface by spectral coefficients such as $R_{mn}(s)$ and $Z_{mn}(s)$. Depending on formulation, it also solves for angular mappings, magnetic-field components, current-related flux functions, and the rotational-transform profile.

Typical fixed-boundary inputs include:

- plasma boundary Fourier coefficients;
- toroidal magnetic flux;
- pressure profile;
- either current information or rotational-transform constraints;
- radial and angular resolution;

- symmetry and sign conventions.

Typical free-boundary inputs additionally include coil geometry, coil currents, conducting structures, and vacuum-region settings.

18.2 Variational principle

Many ideal-MHD equilibrium methods use an energy principle. A representative total energy is

$$W = \int_V \left[\frac{B^2}{2\mu_0} + \frac{p}{\gamma_{\mathrm{ad}} - 1} \right] dV, \tag{18.1}$$

subject to constraints such as conservation of magnetic flux and an equation of state. A stationary point of this constrained functional satisfies ideal-MHD force balance.

The phrase **variational** means that the solver changes the surface representation to reduce a residual or energy gradient derived from W. It does not mean that every stationary point is automatically stable; the second variation is relevant to stability.

18.3 VMEC concept

The Variational Moments Equilibrium Code, abbreviated VMEC, represents nested flux surfaces spectrally and seeks a three-dimensional ideal-MHD equilibrium by minimizing force residuals associated with a variational formulation. The radial direction is discretized, and angular dependence is represented using Fourier series.

Important modeling assumptions include:

1. nested toroidal flux surfaces are imposed;
2. pressure is a flux function;
3. magnetic islands and stochastic regions are not directly represented;
4. convergence depends on radial and spectral resolution, initial guess, profiles, and boundary shape.

VMEC output is often transformed into Boozer coordinates using a separate post-processing step because the coordinates used internally by the equilibrium solver are not automatically the coordinates best suited for orbit and neoclassical calculations. The foundational variational moment method is described by Hirshman and Whitson [7].

18.4 Alternative equilibrium formulations

Other formulations include force-balance Newton solvers, near-axis expansions, free-boundary integral methods, and multi-region relaxed MHD. Multi-region models can represent interfaces between relaxed plasma volumes and can accommodate magnetic islands or stochastic regions more naturally than a globally nested-surface model [8]. The appropriate formulation depends on the physics question.

18.5 Discretization

A typical numerical representation uses:

- a radial grid s_j, with index $j = 0, \ldots, N_s - 1$ for N_s stored radial points;
- Fourier modes (m, n) up to selected truncations;
- numerical quadrature in the angular directions;
- finite differences, finite elements, splines, or spectral bases in radius.

The unknown coefficient vector may be written abstractly as

$$\mathbf{a} = \{R_{mn}(s_j), Z_{mn}(s_j), \ldots\}. \tag{18.2}$$

The discrete nonlinear equilibrium equations are

$$\mathbf{F}(\mathbf{a}) = \mathbf{0}, \tag{18.3}$$

where $\mathbf{F}$ is a vector of force-balance and constraint residuals.

18.6 Nonlinear iteration

A Newton method updates

$$\mathbf{a}^{(k+1)} = \mathbf{a}^{(k)} + \delta\mathbf{a}^{(k)}, \tag{18.4}$$

where

$$\mathbf{J}_F(\mathbf{a}^{(k)})\delta\mathbf{a}^{(k)} = -\mathbf{F}(\mathbf{a}^{(k)}), \tag{18.5}$$

and $\mathbf{J}_F$ is the Jacobian matrix of the residual. Variational descent methods instead move along a preconditioned negative energy gradient. Practical codes use damping, preconditioning, continuation in pressure or boundary deformation, and problem-specific regularization.

18.7 Continuation

A difficult finite-beta equilibrium may be reached gradually. Introduce a continuation parameter $0 \le \lambda \le 1$ and solve a sequence

$$p_\lambda(s) = \lambda p_{\mathrm{target}}(s). \tag{18.6}$$

The converged solution at one λ becomes the initial guess for the next. Continuation can also be applied to boundary shaping, plasma current, or coil perturbations.

18.8 Convergence criteria

A solver should not declare convergence from iteration count alone. Useful criteria include:

$$\epsilon_F = \frac{\|\nabla p - \mathbf{J} \times \mathbf{B}\|}{F_{\mathrm{ref}}}, \tag{18.7}$$

$$\epsilon_B = \frac{\|\nabla \cdot \mathbf{B}\|}{B_{\mathrm{ref}}/L_{\mathrm{ref}}}, \tag{18.8}$$

$$\epsilon_a = \frac{\|\mathbf{a}^{(k+1)} - \mathbf{a}^{(k)}\|}{\|\mathbf{a}^{(k)}\| + a_{\mathrm{floor}}}. \tag{18.9}$$

Here F_{ref}, B_{ref}, L_{ref}, and a_{floor} are explicitly chosen scaling quantities. Norm type, quadrature weighting, and excluded singular points must be documented.

18.9 Force residuals in curvilinear coordinates

A small scalar residual does not guarantee that every physical component is small. The force residual

$$\mathbf{f} = \mathbf{J} \times \mathbf{B} - \nabla p \tag{18.10}$$

should be examined in radial, poloidal, and toroidal directions, or decomposed into components parallel and perpendicular to $\mathbf{B}$. Surface-localized residuals can be hidden by a global average.

18.10 Resolution studies

Convergence must be checked with respect to:

- radial grid size N_s;
- maximum poloidal mode $m_{\max}$;
- maximum toroidal mode $n_{\max}$;
- angular quadrature grid;

- nonlinear tolerance.

A geometry quantity Q should approach a stable value as resolution increases. A useful relative difference is

$$\delta_Q^{(h)} = \frac{|Q_h - Q_{h/2}|}{|Q_{h/2}| + Q_{\mathrm{floor}}}, \tag{18.11}$$

where h denotes an effective discretization scale and Q_{floor} prevents division by a negligible number.

18.11 Failure modes

Common equilibrium-solver failures include:

- self-intersecting or poorly parameterized surfaces;
- loss of nested surfaces in the physical configuration;
- insufficient spectral resolution;
- excessive pressure gradient or incompatible profiles;
- coordinate singularities and negative Jacobian values;
- convergence to an unintended branch;
- sensitivity to initial guess;
- free-boundary mismatch between plasma and vacuum fields.

A failed solve must produce explicit flags and diagnostics rather than silently exporting geometry.

19 Coordinate transformation and interpolation

19.1 Why transformations are unavoidable

Different calculations prefer different representations:

- coil and wall geometry are naturally expressed in Cartesian or cylindrical coordinates;
- equilibrium solvers use internal flux coordinates and spectral coefficients;
- guiding-center solvers often prefer Boozer coordinates or direct Cartesian fields;
- wave solvers require metric coefficients and parallel derivatives;
- transport solvers frequently use a one-dimensional normalized flux coordinate.

Module 1 must therefore define transformations, not only raw field values.

19.2 Forward and inverse mappings

The forward mapping

$$(s, \theta, \zeta) \mapsto \mathbf{x} \tag{19.1}$$

is evaluated from the surface representation. The inverse mapping

$$\mathbf{x} \mapsto (s, \theta, \zeta) \tag{19.2}$$

is generally nonlinear and may require root finding. The inverse can be nonunique outside the nested-surface region or near self-intersections, so its domain must be restricted.

A Newton inverse solves

$$\mathbf{r}(s, \theta, \zeta) = \mathbf{x}(s, \theta, \zeta) - \mathbf{x}_{\text{target}} = \mathbf{0}. \tag{19.3}$$

The update uses the coordinate-basis matrix

$$\begin{bmatrix} \partial_s \mathbf{x} & \partial_\theta \mathbf{x} & \partial_\zeta \mathbf{x} \end{bmatrix}. \tag{19.4}$$

Near a coordinate singularity or a poorly shaped surface, this matrix can be ill-conditioned.

19.3 Interpolation requirements

A downstream solver may evaluate geometry at points not present on the equilibrium grid. Interpolation should preserve:

- periodicity in angular coordinates;
- continuity across field-period boundaries;
- regularity at the magnetic axis;
- stated order of accuracy;
- consistency between a field and its derivatives.

Interpolating B, ∇B, and $\boldsymbol{\kappa}$ independently can violate derivative identities. A differentiable spectral representation is preferable when practical.

19.4 Conservative remapping of flux functions

When a profile such as density or pressure is transferred from one radial grid to another, pointwise interpolation does not necessarily conserve integrated particle number or energy. A conservative remap should preserve an integral such as

$$\mathcal{N} = \int_0^1 n(s)V'(s)\,ds. \tag{19.5}$$

For cell averages, this is accomplished by matching integrals over overlapping radial cells rather than simply matching values at cell centers.

20 Module 1 output contract

20.1 Minimum geometry product

A useful equilibrium data product must identify the equilibrium uniquely and contain enough information for independent checks. At minimum, it should include:

1. coordinate definitions and angle orientation;
2. unit system and normalization constants;
3. field-period number N_{fp};
4. magnetic-axis representation;
5. surface mapping $\mathbf{x}(s, \theta, \zeta)$ or equivalent Fourier coefficients;
6. magnetic field $\mathbf{B}$ and magnitude B;
7. rotational transform $\iota(s)$;
8. pressure $p(s)$ and, when used, mass density $\rho_m(s)$;
9. Jacobian $\sqrt{g}$ and metric tensors or enough information to reconstruct them;
10. enclosed volume $V(s)$ and $V'(s)$;
11. derivatives required for ∇B, curvature, and parallel operators;
12. boundary and last-closed-surface definition;
13. convergence diagnostics and failure flags.

20.2 Recommended metadata

Recommended metadata include:

- solver name and version;

- source input-file checksum;
- fixed-boundary or free-boundary designation;
- pressure and current-profile formulas;
- radial and spectral resolution;
- nonlinear tolerance and achieved residuals;
- sign convention for toroidal flux, poloidal flux, ι, and angles;
- whether ζ is geometric or magnetic toroidal angle;
- coordinate transformation method;
- date and provenance of coil geometry;
- known warnings, extrapolation regions, and invalid domains.

20.3 Suggested array shapes

For a sampled grid with N_s radial points, N_θ poloidal points, and N_ζ toroidal points, representative arrays are

$$\begin{aligned}
\texttt{position} &: [N_s, N_\theta, N_\zeta, 3], && (20.1)\\
\texttt{B_vector} &: [N_s, N_\theta, N_\zeta, 3], && (20.2)\\
\texttt{B_magnitude} &: [N_s, N_\theta, N_\zeta], && (20.3)\\
\texttt{jacobian} &: [N_s, N_\theta, N_\zeta], && (20.4)\\
\texttt{iota} &: [N_s], && (20.5)\\
\texttt{pressure} &: [N_s], && (20.6)\\
\texttt{Vprime} &: [N_s]. && (20.7)
\end{aligned}$$

The final software schema may use spectral coefficients instead of dense arrays. Whatever representation is chosen, axis order and memory layout must be documented.

20.4 Downstream consumers

The main consumers are:

Module	Required Module 1 quantities	Physical use
Module 2	s, volume, profiles, field strength	Places energetic-particle sources and distributions on the equilibrium.
Module 3	$\mathbf{B}$, B, ∇B, $\boldsymbol{\kappa}$, coordinates, wall mapping	Advances guiding centers and identifies confined and lost trajectories.
Module 4	ι, B, ρ_m, metric, $\mathbf{b} \cdot \nabla$	Computes shear-Alfven continua, harmonic coupling, and eigenmodes.
Module 5	mode geometry plus orbit frequencies derived from Module 1	Evaluates wave-particle resonances and energetic-particle drive.
Module 6	V', radial coordinate, mode geometry	Evolves reduced fast-ion transport conservatively.
Module 7	V', surface areas, density and field data	Evolves heating and thermal profiles.
Module 8	all typed state products and provenance	Coordinates multiphysics exchanges and detects stale geometry.
Module 9	parameterized equilibrium inputs and physical residuals	Trains and validates equilibrium or downstream surrogate models.
Module 10	viewing geometry and field mapping	Generates synthetic diagnostic signals.
Module 11	named physical dependencies and interventions	Learns candidate coupling graphs without treating arrays as anonymous features.

Table 3: Principal equilibrium dependencies across the project.

20.5 Versioning and immutable state products

An equilibrium should be treated as an immutable state product with a unique identifier. If pressure, coil current, boundary shape, or plasma current changes enough to require a new solve, the resulting geometry receives a new identifier. Downstream outputs should record the exact equilibrium identifier used. This prevents a mode spectrum or orbit-loss map from being paired accidentally with a different geometry.

21 Worked conceptual examples

21.1 Example 1: pressure as a flux function

Suppose a nested magnetic surface is labeled by s. Start from force balance,

$$\nabla p = \mathbf{J} \times \mathbf{B}. \tag{21.1}$$

Dot with $\mathbf{B}$:

$$\mathbf{B} \cdot \nabla p = 0. \tag{21.2}$$

Parameterize a field line by τ using $d\mathbf{x}/d\tau = \mathbf{B}$. The pressure derivative along the line is

$$\frac{dp}{d\tau} = \frac{d\mathbf{x}}{d\tau} \cdot \nabla p = \mathbf{B} \cdot \nabla p = 0. \tag{21.3}$$

Therefore p is constant along that field line. If the line densely covers the surface and p is continuous, all points on the surface have the same pressure, so $p = p(s)$.

21.2 Example 2: field-line closure

Let $\iota = 2/5$ and use the straight-line solution

$$\theta(\zeta) = \theta_0 + \frac{2}{5}\zeta. \tag{21.4}$$

After five toroidal turns, $\Delta\zeta = 10\pi$. The poloidal advance is

$$\Delta\theta = \frac{2}{5}(10\pi) = 4\pi, \tag{21.5}$$

which is two poloidal turns. The angular coordinates return to their starting values modulo 2π, so the ideal field line closes.

21.3 Example 3: a rational Alfven resonance surface

Take a mode with $(m, n) = (2, 1)$ and the transform profile

$$\iota(s) = 0.35 + 0.30s. \tag{21.6}$$

The simple parallel wave number vanishes when

$$2\iota(s) - 1 = 0. \tag{21.7}$$

Thus

$$2(0.35 + 0.30s) - 1 = 0 \quad \Rightarrow \quad s = 0.5. \tag{21.8}$$

The surface $s = 0.5$ is the rational surface $\iota = 1/2$ for this harmonic. In a more complete shear-Alfven model, coupling, compressibility, and finite-frequency effects determine the actual continuum behavior near this location.

21.4 Example 4: beta estimate

Let $p = 2.0 \times 10^5$ Pa and $B = 2.5$ T. The magnetic pressure is

$$\frac{B^2}{2\mu_0} = \frac{(2.5)^2}{2(4\pi \times 10^{-7})} \approx 2.49 \times 10^6 \text{ Pa}. \tag{21.9}$$

The local beta is

$$\beta \approx \frac{2.0 \times 10^5}{2.49 \times 10^6} \approx 0.080, \tag{21.10}$$

or approximately 8.0%. This does not mean magnetic forces exceed pressure forces everywhere by exactly a factor of 12.5; equilibrium depends on gradients and curvature, not only local energy-density ratios.

21.5 Example 5: field-period Fourier variation

Suppose $N_{\text{fp}} = 4$ and a surface contains the term

$$R_{1,1}(s)\cos(\theta - 4\zeta). \tag{21.11}$$

Increasing ζ by one field period, $\Delta\zeta = 2\pi/4$, changes the phase by

$$-4\Delta\zeta = -2\pi. \tag{21.12}$$

The cosine is unchanged, confirming fourfold toroidal periodicity.

22 Verification and validation

22.1 Definitions

Verification asks whether the equations were solved correctly. **Validation** asks whether those equations and inputs adequately represent the physical system for the intended use. A solver can be verified but invalid for a given experiment, or physically appropriate but implemented incorrectly.

22.2 Code verification tests

A Module 1 implementation should include at least the following tests.

22.2.1 Dimensional and convention tests

Every exported quantity must carry units or an explicit normalization. Tests should check angle ranges, coordinate orientation, flux signs, and the relation $q = 1/\iota$ under the project convention.

22.2.2 Divergence-free test

Evaluate

$$\epsilon_{\nabla\cdot B} = \frac{L_{\mathrm{ref}}\|\nabla\cdot\mathbf{B}\|_2}{\|\mathbf{B}\|_2}. \tag{22.1}$$

The norm should decrease with increased numerical resolution until limited by solver tolerance or interpolation error.

22.2.3 Surface-tangency test

Evaluate

$$\epsilon_{B\cdot\nabla s} = \frac{\|\mathbf{B}\cdot\nabla s\|_2}{\|\mathbf{B}\|_\infty\|\nabla s\|_2 + \epsilon_{\mathrm{floor}}}. \tag{22.2}$$

A nested-surface equilibrium should make this small throughout the valid domain.

22.2.4 Force-balance test

Evaluate

$$\epsilon_{\mathrm{force}} = \frac{\|\mathbf{J}\times\mathbf{B} - \nabla p\|_2}{\|\nabla p\|_2 + \|\mathbf{J}\times\mathbf{B}\|_2 + \epsilon_{\mathrm{floor}}}. \tag{22.3}$$

The test should also inspect maximum and surface-resolved residuals.

22.2.5 Field-line transform test

Trace field lines independently and estimate

$$\iota_{\mathrm{trace}}(s) = \frac{\Delta\theta}{\Delta\zeta} \tag{22.4}$$

over many turns. Compare with the exported $\iota(s)$. The discrepancy should decrease with tracing tolerance and interpolation resolution.

22.2.6 Flux conservation test

Compute magnetic flux through equivalent cross-sections of the same flux tube. Because $\nabla\cdot\mathbf{B} = 0$, the flux should be independent of cross-section up to numerical error.

22.2.7 Coordinate round-trip test

For points inside the valid domain, apply

$$(s, \theta, \zeta) \to \mathbf{x} \to (\tilde{s}, \tilde{\theta}, \tilde{\zeta}). \tag{22.5}$$

Compare s directly and compare angles modulo 2π. Failures near the axis should be treated with coordinate-aware tolerances.

22.2.8 Manufactured circular-torus test

Recover the analytic mappings, volume, V', and field-line solution of Section 17. This separates geometry-interface errors from the complexity of a full equilibrium code.

22.3 Solution verification

Even a verified code requires case-specific convergence studies. For each equilibrium, report changes in quantities such as

- magnetic-axis location;
- $\iota(s)$;
- volume-averaged beta;
- minimum Jacobian;
- selected B_{mn} coefficients;
- $B_{\min}$ and $B_{\max}$;
- force residual;
- derived Module 4 continuum frequencies.

The last item is a cross-module sensitivity test: a geometry may appear visually converged while small transform or metric errors still shift resonant frequencies.

22.4 Validation sources

Possible validation sources include:

- comparison with an independently implemented equilibrium code;

- comparison with published benchmark equilibria;
- magnetic-diagnostic reconstruction from an experiment;
- pressure-profile and plasma-boundary measurements;
- vacuum-field mapping before plasma operation;
- cross-checks with coil-current perturbation experiments.

Validation must be tied to an intended prediction. Agreement in boundary shape does not automatically validate energetic-particle orbit confinement.

22.5 Uncertainty sources

Important uncertainties include:

- coil-position and coil-current errors;
- plasma-pressure and current-profile uncertainty;
- diagnostic reconstruction error;
- finite spectral and radial resolution;
- coordinate-transformation error;
- model-form error from scalar-pressure ideal MHD;
- omitted islands, stochasticity, flow, or anisotropy.

Uncertainty should be propagated to quantities actually used by downstream modules, such as ι, orbit-loss fraction, or eigenfrequency.

23 Equilibrium surrogates and physics constraints

23.1 Why build a surrogate?

A high-fidelity equilibrium solve inside a large parameter scan, control loop, uncertainty analysis, or digital twin can be expensive. A surrogate seeks a rapid approximation

$$\mathcal{G} : (\text{boundary, coils, profiles, controls}) \mapsto (\mathbf{B}, \iota, \text{geometry, diagnostics})\,. \tag{23.1}$$

Possible models include neural operators, reduced-basis methods, Gaussian processes, and physics-informed neural networks.

23.2 Physical constraints

A useful equilibrium surrogate should be checked against

$$\nabla \cdot \mathbf{B} = 0, \tag{23.2}$$

$$\mathbf{B} \cdot \nabla s = 0, \tag{23.3}$$

$$\mathbf{J} - \frac{1}{\mu_0} \nabla \times \mathbf{B} = 0, \tag{23.4}$$

$$\mathbf{J} \times \mathbf{B} - \nabla p = 0, \tag{23.5}$$

in the domain where these equations apply. It should also respect periodicity, stellarator symmetry when imposed, boundary conditions, and positive Jacobian orientation.

23.3 Representation choices

Predicting Cartesian components of $\mathbf{B}$ independently can violate $\nabla \cdot \mathbf{B} = 0$. Better choices may include:

- predicting a vector potential $\mathbf{A}$ and setting $\mathbf{B} = \nabla \times \mathbf{A}$;
- predicting flux-surface coordinates and using a divergence-free contravariant representation;
- predicting spectral coefficients constrained by symmetry;
- learning corrections to a reduced conventional equilibrium rather than the full field.

Each representation moves some physical constraints from the loss function into the architecture.

23.4 Validation requirements

Low average data error is insufficient. A surrogate should be tested for:

1. in-distribution interpolation;
2. out-of-distribution boundary and profile changes;
3. force and divergence residuals;
4. topology preservation and positive Jacobian;
5. accuracy of derived quantities such as ι, $k_\parallel$, curvature, and orbit invariants;
6. uncertainty calibration;
7. graceful failure or rejection outside the training domain.

23.5 Do not replace the reference solver

The conventional equilibrium solver remains the reference for generating training data, checking residuals, and identifying surrogate failure. The surrogate accelerates selected workflows; it does not eliminate the need for conventional physics verification.

24 Limitations of the baseline Module 1 model

The textbook description and planned interface are broader than the current executable repository capability. The baseline does not yet implement a validated three-dimensional equilibrium solver. Furthermore, the primary physical model described here has the following limitations unless explicitly extended:

1. scalar, isotropic pressure rather than an anisotropic pressure tensor;
2. static equilibrium without strong bulk flow;
3. ideal MHD rather than resistive, two-fluid, or kinetic equilibrium;
4. globally nested flux surfaces;
5. no self-consistent magnetic islands or stochastic regions;
6. no scrape-off-layer or open-field-line equilibrium;
7. no direct engineering stress, thermal, or structural coil calculation;
8. no guarantee of MHD stability or good particle confinement;
9. no automatic electrostatic-potential solution;
10. no uncertainty-calibrated experimental reconstruction.

These boundaries are not defects when stated clearly. They define the domain in which Module 1 outputs may be trusted.

25 Conceptual map for interview-level explanation

A concise but technically correct explanation is:

A stellarator equilibrium is a three-dimensional arrangement of magnetic field, plasma current, and pressure that satisfies $\nabla p = \mathbf{J} \times \mathbf{B}$, $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$, and $\nabla \cdot \mathbf{B} = 0$. The field is organized into nested toroidal magnetic surfaces. Field lines wind poloidally and toroidally on each surface, with average pitch measured by the rotational transform ι. Surface shape, field-strength variation, metric coefficients, curvature, and ι determine particle orbits, Alfven-wave propagation, resonances, and transport. A stellarator obtains this geometry primarily from three-dimensional external coils, so equilibrium computation and coordinate transformation are foundational inputs to every later kinetic-MHD module.

The following distinctions should remain explicit:

- equilibrium versus stability;
- vacuum field versus finite-pressure plasma equilibrium;
- geometric angles versus straight-field-line angles;
- magnetic surface label versus physical radius;
- rotational transform ι versus safety factor q;
- accurate field values versus accurate derivatives;
- a nested-surface representation versus a field containing islands or stochasticity;
- a surrogate prediction versus a verified conventional solution.

26 Chapter summary

1. Magnetic confinement exploits rapid gyromotion and relatively free motion along magnetic field lines. Closing field lines toroidally avoids simple end loss.
2. A stellarator uses three-dimensional external magnetic geometry to produce field-line twist without relying on a large transformer-driven plasma current.
3. Static ideal-MHD equilibrium satisfies

$$\nabla p = \mathbf{J} \times \mathbf{B}, \qquad \nabla \times \mathbf{B} = \mu_0 \mathbf{J}, \qquad \nabla \cdot \mathbf{B} = 0.$$

4. Force balance implies $\mathbf{B} \cdot \nabla p = 0$, so pressure is a flux function when nested surfaces exist.
5. A magnetic surface satisfies $\mathbf{B} \cdot \nabla s = 0$. The normalized toroidal-flux coordinate s labels nested surfaces, while $\rho = \sqrt{s}$ is a convenient radius-like coordinate.

6. Rotational transform $\iota = d\theta/d\zeta$ measures average field-line pitch. The safety factor is $q = 1/\iota$ under the project convention.
7. Radial variation of ι produces magnetic shear. Rational values $\iota = n/m$ identify surfaces important for islands and wave resonances.
8. Straight-field-line coordinates make field lines straight in angular coordinate space. Boozer coordinates are especially useful for particle-orbit and quasisymmetry analysis.
9. Three-dimensional surfaces and field strength are represented efficiently by Fourier series in poloidal and toroidal angles.
10. The vacuum coil field and plasma-current field both contribute to the operating equilibrium.
11. Fixed-boundary calculations prescribe the plasma boundary; free-boundary calculations solve the plasma-vacuum interface with coil fields.
12. A numerical equilibrium is a nonlinear constrained problem requiring radial and spectral convergence studies, force-balance checks, and coordinate-quality diagnostics.
13. Downstream modules require not only $\mathbf{B}$ but also B, derivatives, curvature, metrics, Jacobian, ι, V', profiles, conventions, provenance, and uncertainty information.
14. A physics-informed surrogate must preserve equilibrium constraints and derived-quantity accuracy; it cannot be validated by field-value error alone.

A Vector identities used in this chapter

A.1 Triple-product identities

For vectors $\mathbf{A}$, $\mathbf{C}$, and $\mathbf{D}$,

$$\mathbf{A} \cdot (\mathbf{C} \times \mathbf{D}) = \mathbf{C} \cdot (\mathbf{D} \times \mathbf{A}) = \mathbf{D} \cdot (\mathbf{A} \times \mathbf{C}), \tag{A.1}$$

$$\mathbf{A} \times (\mathbf{C} \times \mathbf{D}) = \mathbf{C}(\mathbf{A} \cdot \mathbf{D}) - \mathbf{D}(\mathbf{A} \cdot \mathbf{C}). \tag{A.2}$$

A.2 Magnetic-force identity

For a differentiable vector field $\mathbf{B}$,

$$(\nabla \times \mathbf{B}) \times \mathbf{B} = (\mathbf{B} \cdot \nabla)\mathbf{B} - \frac{1}{2}\nabla B^2 - \mathbf{B}(\nabla \cdot \mathbf{B}). \tag{A.3}$$

When $\nabla \cdot \mathbf{B} = 0$, this yields Equation (3.15).

A.3 Divergence of a curl

For a sufficiently smooth vector potential $\mathbf{A}$,

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0. \tag{A.4}$$

Thus representing $\mathbf{B} = \nabla \times \mathbf{A}$ enforces the solenoidal condition analytically.

B Derivation of the straight-field-line representation

Let ψ label magnetic surfaces and let θ and ζ be periodic angles. A field tangent to surfaces can be represented using cross products containing $\nabla\psi$:

$$\mathbf{B} = A(\psi, \theta, \zeta)\nabla\psi \times \nabla\theta + C(\psi, \theta, \zeta)\nabla\zeta \times \nabla\psi. \tag{B.1}$$

The two terms are tangent to constant-ψ surfaces. A suitable angular-coordinate choice and flux normalization set $A = 1$ and make $C = \iota(\psi)$, giving Equation (7.11). This coordinate construction is nontrivial globally; the equation should be understood as the defining representation of an appropriate straight-field-line system, not as a statement that arbitrary geometric angles already have this property.

C Example project interface checklist

Before a Module 1 state product is accepted by another module, verify:

1. The equilibrium identifier and source checksum are present.
2. SI units or normalization constants are present for every dimensional field.
3. The meanings of s, ρ, θ, and ζ are documented.
4. Angle orientation, flux signs, and the q–ι convention are documented.
5. The valid domain and last closed flux surface are identified.
6. The field-period convention and Fourier convention are identified.
7. $\mathbf{B}$, B, position, Jacobian, metric, and ι are mutually consistent.
8. Derivatives are generated from the same representation as the primary field where possible.
9. The minimum Jacobian is positive under the stored orientation.
10. Force, divergence, tangency, and transform residuals pass specified tolerances.

11. Radial and spectral convergence evidence is attached.
12. Interpolation and extrapolation behavior is documented.
13. The geometry does not silently extrapolate outside the valid plasma or vacuum domain.
14. Downstream outputs retain the equilibrium identifier.

D Review questions and exercises

1. Explain why a purely toroidal magnetic field is not generally sufficient for good toroidal confinement. Distinguish particle streaming from cross-field drifts.
2. Starting from $\nabla p = \mathbf{J} \times \mathbf{B}$, prove that both $\mathbf{B}$ and $\mathbf{J}$ are tangent to constant-pressure surfaces.
3. Derive Equation (3.13) for the perpendicular current.
4. Use the magnetic-force identity to interpret magnetic pressure and tension.
5. For $B = 3$ T and $p = 1.5 \times 10^5$ Pa, calculate the local beta.
6. A field line has $\iota = 0.42$. How many poloidal turns does it make during ten toroidal turns? Is the line guaranteed to close?
7. For $\iota(s) = 0.3 + 0.4s$, locate the surfaces with $\iota = 1/2$ and $\iota = 2/3$, if they lie in $0 \leq s \leq 1$.
8. Show that Equation (7.11) gives $d\theta/d\zeta = \iota$.
9. Explain why $\rho = \sqrt{s}$ should not automatically be interpreted as geometric distance from the magnetic axis.
10. Describe the difference between fixed-boundary and free-boundary equilibrium.
11. Explain why a vacuum coil field is not necessarily the same as a finite-beta plasma equilibrium.
12. What physics is excluded when globally nested surfaces are imposed?
13. Why can accurate interpolation of B still produce inaccurate guiding-center orbits if ∇B is noisy?
14. Write the Fourier phase for a mode with poloidal number $m = 3$, toroidal number $n = 2$, and $N_{\text{fp}} = 4$ under the convention in Equation (9.1).
15. Explain the distinction between stellarator symmetry and quasisymmetry.
16. Give three quantities from Module 1 that Module 4 needs and explain why.

17. Construct a convergence table for $\iota(0.5)$ as $m_{\max}$, $n_{\max}$, and N_s are increased. What pattern would support numerical convergence?

18. Propose a divergence-free neural representation for $\mathbf{B}$ and identify one disadvantage.

19. Explain why a low mean-square error in B does not guarantee a useful equilibrium surrogate.

20. State the most important assumptions that must accompany any claim based on this Module 1 model.

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